

A continuous random variable

A continuous random variable can take values throughout an interval.

Would exact file-transfer duration be discrete or continuous?

A probability density

A continuous probability density function (PDF), written $f(x)$, is nonnegative and has total area one.

What would fail if part of a proposed density lay below the horizontal axis?

The entire distribution

Model: a rectangular density extends from 0 to 10 seconds.

What height makes its total area equal to one?

Probability as area

Model: $f(x) = 0.1$ for $0 \leq x \leq 10$ seconds, zero elsewhere. Interval probabilities are areas under the density.

For the rectangular density, what region represents durations between 3 and 7 seconds?

An interval probability

Model: the density is 0.1 on the interval from 0 to 10 seconds and zero elsewhere.

What is the probability of a duration between 3 and 7 seconds?

Graphing a continuous density

Model: $f(x) = 0.1$ for $0 \leq x \leq 10$ seconds, zero elsewhere.

How would you label both axes and shade the area representing $P(3 < X < 7)$?

Height is not probability

Model: a uniform density has height 2 on the interval from 0 to 0.5 seconds.

Can a density exceed one without a probability exceeding one?

One exact value

A single value has probability zero under a continuous density.

Does probability zero mean that a measured duration cannot be reported as exactly 4 seconds?

Rounded measurements

Model: an exact duration is uniform from 0 to 10 seconds and is rounded to the nearest second.

What interval of exact durations would be reported as 4 seconds?

Endpoints

Model: an exact duration has a continuous density.

Would including an interval endpoint change its probability?

The cumulative distribution

The cumulative distribution function (CDF), $F(x)$, gives the probability at or below a cutoff.

$$F(x) = P(X \leq x)$$

What does $F(6)$ describe when X measures a duration in seconds?

A cumulative probability

Model: X is uniform from 0 to 10 seconds.

What is $F(6)$?

Beyond a cutoff

Model: a continuous duration has $F(6)=0.72$.

What is the probability that the duration exceeds 6 seconds?

Between two cutoffs

Model: a continuous duration has $F(3)=0.20$ and $F(7)=0.85$.

What is the probability that the duration lies between 3 and 7 seconds?

Outside an interval

Model: a continuous duration falls between 3 and 7 seconds with probability 0.65.

What probability belongs outside that interval?

A possible cumulative function

A cumulative distribution function never decreases as the cutoff increases.

Could $F(4)=0.6$ and $F(8)=0.4$ describe the same distribution?

A percentile

The 80th percentile is a cutoff with 80% of the distribution at or below it.

If a continuous duration has $F(9)=0.80$, what percentage exceeds 9 seconds?

Choosing the probability

A transfer must finish within 8 seconds to meet a deadline.

Which probability describes meeting the deadline: $F(8)$, $1-F(8)$, or the density at 8?